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VEL SOFTWARE SOLUTIONS · INTERNAL OPERATING SYSTEM

SaaS & Web Development Company Operating System

From market research to product development, production, sales and recurring revenue – employee-ready SOPs, templates, financial models, checklists and approval gates.

Version 1.1 | Fees verified 2026-11-03 | Generated 2026-11-03 | 12 approval gates | 14 stages | 20 templates | Worked example included

Read first
Figures marked ASSUMPTION are planning placeholders, not market or provider facts. No competitor, price or domain availability has been invented. Replace assumptions with cited, dated sources before making commercial decisions. Legal, tax and compliance items need review by qualified professionals.

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Company intake & assumptions

Intake question	Working default (edit me)	Why it matters
Company name	VEL Software Solutions (owner and developer: V.LOKESHWARAN)	Legal entity name for contracts, registrar and billing
Business model	B2B SaaS subscriptions + custom web development (project work)	Recurring revenue is the long-term goal; project work funds the runway
Employee count	1 founder + 5 roles (Tech Lead, Full-stack dev, UI/UX, QA, DevOps) – can be part-time	Roles can be combined; gates cannot be skipped
Target customers	Small and mid-sized businesses (10–200 staff) in India; service-led businesses first	Narrow ICP gives cheaper acquisition
Available budget	[Fill in] – model uses ₹10–15 lakh first-product budget as a planning figure	ASSUMPTION
Technology preferences	TypeScript, React/Next.js, Node.js or FastAPI, PostgreSQL, managed PaaS	Final choice only at Gate G3
Initial product focus	Sample: SlotWise – appointment scheduling and reminders for small clinics	Illustrative; replace with your first validated opportunity

How to use this page

The defaults above let the framework and the worked example run end-to-end. Replace them with your real answers; the financial model recalculates from the same inputs (see the web calculator). Nothing here is a fact about your company until you confirm it.

Company-wide operating framework

Lifecycle

1. Market research
2. Customer validation
3. Product definition
4. Input/Output freeze
5. Architecture
6. UI/UX
7. VS Code & GitHub
8. Development
9. Testing
10. Production cost
11. Pricing approval
12. Domain & DNS
13. Deployment
14. Customer showcase
15. Sales
16. Handover
17. Maintenance
18. Recurring revenue

Approval gates

Gate	Name	Evidence required	Approver
G1	Opportunity Approval	Market Research & Opportunity Report	Business Owner
G2	Scope Freeze	PRD + I/O Spec + Scope Freeze	Product Owner + Customer
G3	Architecture Approval	Architecture Document	Tech Lead + Business Owner
G4	Design Approval	UI/UX system + prototype	Product Owner + Customer
G5	Repo Readiness	Configured repository + CI	Tech Lead
G6	Release Candidate	Tested software, QA + security reports	Product Owner + Tech Lead
G7	Cost Model Approval	Production cost model	Business Owner + Finance
G8	Pricing Approval	Pricing & profitability model	Business Owner
G9	Release Approval	Domain/DNS + Deployment reports	Tech Lead + Business Owner
G10	Commercial Approval	Proposal/SOW/contract	Business Owner
G11	Operational Readiness	Maintenance agreement + runbooks	Tech Lead
G12	Handover & Renewal	Acceptance form + handover pack	Customer + Support Lead

Responsibility matrix

Activity	Owner (Founder)	Product Owner	Tech Lead	Developer	Designer	QA	DevOps
Market research	A	R	C	–	–	–	–
PRD / Scope	A	R	C	C	C	C	–
Architecture	C	C	A/R	C	–	–	C
UI/UX	C	A	C	C	R	C	–
Repo / CI	–	–	A	C	–	–	R
Development	–	C	A	R	C	C	C
Testing	–	C	C	C	–	A/R	–
Security review	C	–	A/R	C	–	C	C
Cost & pricing	A	R	C	–	–	–	C
Deployment	C	–	A	C	–	C	R
Sales & delivery	A	R	C	–	–	–	–
Maintenance	C	C	A	R	–	C	R

R = Responsible, A = Accountable, C = Consulted, – = not involved.

Critical rules (non-negotiable)

1. Never start coding before product scope approval (G2).
2. Never launch without testing and security review.
3. Never estimate production cost without documenting usage assumptions.
4. Never confuse one-time development cost with monthly operating cost.
5. Never treat free-tier services as guaranteed permanent infrastructure.
6. Never invent domain availability or DNS values.
7. Never calculate profit using revenue alone.
8. Never hide recurring third-party costs.

9. Never ignore maintenance and customer-support labour.
10. Never recommend an expensive architecture without justification.
11. Never claim a product is production-ready without evidence.
12. Never proceed past an approval gate without authorisation.

Reading a stage

Every stage card has the same ten parts: instructions, responsibilities, tools, time, cost, deliverables, quality checklist, approval criteria, official sources, common mistakes. Stage 1–11 follow product order; 12–14 are continuous disciplines.

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1 Market Research & Opportunity

G1 – Opportunity Approval

Prevents building a product nobody will pay for. No code is written until the owner signs G1.

1. Detailed instructions

1. Define target market and customer segment; write the problem statement in one sentence (who, what pain, how they cope today).
2. Interview at least 10 target customers (Customer Discovery template). Record willingness to pay as a stated price range, not a yes/no.
3. List 5–10 competitors (direct and indirect). For each: pricing page, feature list, tech stack signals, positioning, reviews (G2, Capterra, Product Hunt, Reddit). Record URL and access date.
4. Build the feature/pricing comparison matrix; extract gaps and the differentiator.
5. Define Ideal Customer Profile (ICP); name buyer, end user and decision-maker (often three different people).
6. Estimate sales cycle and acquisition difficulty (channel, effort, expected CAC range).
7. Size the market bottom-up: TAM = all potential customers × annual contract value; SAM = reachable segment (geography, size, language); SOM = share winnable in 3 years given capacity. Show every assumption.
8. Identify recurring-revenue levers: subscription tiers, usage add-ons, onboarding fee, support plans.
9. Write the Market Research and Product Opportunity Report; submit for G1.

In practice – precise targets and artefacts

- **Timebox:** week 1 desk research (competitors, sizing); weeks 2–3 interviews. Stop at 10 completed interviews or 15 working days, whichever is first.
- **Evidence grading:** tag every finding A (official/primary source or measured), B (credible secondary), C (opinion/assumption). A decision may rest on A/B only.
- **Competitor matrix columns:** name · URL · access date · target segment · plans & prices · free tier/trial · key features · integrations · review score & count · weaknesses quoted from reviews.
- **Pass rule example:** ≥3/10 rank the pain top-3 AND ≥3 accept a paid pilot at or above the break-even price from Stage 8.

2. Employee responsibilities

Role	Responsibility
Product Owner	Owns report and the go/no-go recommendation
Researcher / Analyst	Collects sources, builds comparison matrix
Business Owner	Approves or rejects at G1

3. Required tools

Google Sheets/Excel, G2, Capterra, Product Hunt, Reddit, Google Trends, competitor sites, government/industry reports, interview recording (with consent).

4. Estimated time

2–3 weeks (about 41 h expected for the sample product)

5. Estimated cost

Labour only: ≈ ₹50,000 expected (41 h × ₹1,200). No tooling cost required.

6. Expected deliverables

- Market Research & Product Opportunity Report
- Competitor Analysis sheet (with URLs + access dates)
- Customer Discovery notes (≥10)
- TAM/SAM/SOM workbook

7. Quality checklist

☐ ≥5 competitors compared with source links
☐ Every number has a source or is tagged ASSUMPTION
☐ ≥10 customer conversations logged
☐ ICP, buyer, user, decision-maker named
☐ TAM/SAM/SOM bottom-up with arithmetic shown
☐ Recurring revenue path identified
☐ Risks and kill criteria written

8. Approval criteria

Business Owner signs G1 in the dashboard. Kill criteria: <3 of 10 prospects confirm the pain as top-3 priority, or stated willingness to pay below projected break-even price.

9. Official source references (verify and record access date)

- Google Trends https://trends.google.com/

- G2 https://www.g2.com/
- Capterra https://www.capterra.com/
- Product Hunt https://www.producthunt.com/

10. Common mistakes & prevention

Common mistake	Prevention
Treating friends' enthusiasm as demand	Require stated price + a pilot commitment (LOI or paid pilot)
Top-down TAM ('1% of a huge market')	Use bottom-up count × price
Copying competitor features	Anchor on the validated pain, then differentiate

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2 Product Requirements & Scope Freeze

G2 – Scope Freeze

Defines exactly what will be built; later change is controlled and priced.

1. Detailed instructions

1. Write the PRD: purpose, customers, user roles and permission matrix, core modules, features/sub-features, user journeys.
2. Write the Input–Output Specification: for every input (name, type, unit, required, default, validation, min/max, source, DB mapping) and every output (name, format, calculation, precision, display, export, API shape, error handling).
3. Add non-functional requirements: security, performance targets, mobile responsiveness, browser matrix, backup RPO/RTO, maintenance windows.
4. Write acceptance criteria in Given/When/Then form for every feature; attach at least one sample input→output scenario.
5. Review with customer(s) and engineering; resolve open questions.
6. Freeze: record version, date and approvers. All later changes go through a Change Request.

In practice – precise targets and artefacts

- **Granularity:** a feature is 'specified' only when it has role, trigger, inputs, rules, outputs, error cases and ≥2 Given/When/Then criteria.
- **Measurable NFR examples:** p95 API latency < 500 ms at peak; page LCP < 2.5 s on mid-range mobile; RPO ≤ 24 h, RTO ≤ 4 h; supports last 2 versions of Chrome, Edge, Safari, Firefox.
- **Change-control SLA:** Product Owner answers a CR within 3 working days; approved CRs update the PRD version (v1.0 → v1.1) and the cost model.
- **Freeze evidence:** PDF of PRD + I/O spec with version, date and both signatures stored in the dashboard.

2. Employee responsibilities

Role	Responsibility
Product Owner	Authors PRD, owns acceptance criteria
Tech Lead	Reviews feasibility, estimates, NFRs
QA Lead	Confirms every criterion is testable
Customer / Business Owner	Approves scope

3. Required tools

Docs/Notion/Confluence, Figma (journey maps), spreadsheet for I/O spec, dashboard approval module.

4. Estimated time

2–3 weeks (≈ 78 h)

5. Estimated cost

Labour: ≈ ₹98,500 expected (Product Owner + Tech Lead hours).

6. Expected deliverables

- Approved PRD
- Input–Output Specification
- Scope Freeze Document (versioned)
- Change Request log (empty at freeze)

7. Quality checklist

☐ All 20 PRD headings present
☐ Every input and output specified
☐ Every feature has acceptance criteria
☐ Out-of-scope list explicit
☐ Permissions matrix complete
☐ NFR numbers measurable (e.g. p95 < 500 ms)
☐ Signed by customer and product owner

8. Approval criteria

G2 signed by Product Owner and Customer. A Change Request needs: ID, business justification, technical, cost, timeline and security impact, customer approval and product-owner approval.

9. Official source references (verify and record access date)

- OWASP ASVS (security requirements) https://asvs.owasp.org/
- NIST SP 800-218 SSDF https://csrc.nist.gov/pubs/sp/800/218/final

10. Common mistakes & prevention

Common mistake	Prevention
Vague requirements ('fast', 'easy')	Write measurable targets
Silent scope growth	No work without a CR ID
Missing error-handling spec	Mandatory column in the output spec

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3 Technology Stack & Architecture

G3 – Architecture Approval

Chooses the cheapest architecture that safely meets the requirements, with documented trade-offs.

1. Detailed instructions

1. Draft at least two architecture options (e.g. A: Next.js + Node/TypeScript + PostgreSQL modular monolith on a managed PaaS; B: React SPA + FastAPI + PostgreSQL on a single cloud VM/containers).
2. Score each option 1–5 on: development cost, hosting cost, maintenance cost, scalability, security, performance, developer availability, vendor lock-in, integration fit, long-term operating cost. Weight by project priorities.
3. Default to a modular monolith for the MVP. Microservices require a written justification (independent scaling, team boundaries, compliance isolation).
4. Produce: system diagram, frontend, backend, database architecture, ERD, API documentation (OpenAPI), authentication architecture, threat model (STRIDE), infrastructure diagram, technology justification.
5. Run the threat model against OWASP Top 10 and ASVS level 1 (level 2 for payments/personal data).
6. Tech Lead presents; Business Owner approves cost implications.

In practice – precise targets and artefacts

- **Options table:** score 1–5 × weight (weights sum to 100%). Record the runner-up and why it lost.
- **Module boundaries (modular monolith):** each module owns its tables and exposes a service interface; no cross-module table joins. This keeps a later split possible.
- **Threat model (STRIDE):** list assets, trust boundaries, entry points; for each: Spoofing, Tampering, Repudiation, Information disclosure, DoS, Elevation of privilege → control → test.
- **Multi-tenancy rule:** every tenant-owned row carries tenant_id and every query is filtered server-side; add an automated cross-tenant access test.

2. Employee responsibilities

Role	Responsibility
Tech Lead	Owns the document and the decision
Senior Developer	Peer review and spikes
DevOps Engineer	Infrastructure and cost inputs
Product Owner	Confirms fit to requirements

3. Required tools

draw.io/Excalidraw/Mermaid, dbdiagram.io or pgModeler, OpenAPI editor, threat-model template.

4. Estimated time

1–2 weeks

5. Estimated cost

Labour ≈ ₹92,500 (Tech Lead database/API design hours in sample).

6. Expected deliverables

- Software Architecture Document
- ERD + data dictionary
- OpenAPI draft
- Threat model
- Infrastructure diagram
- Decision log (ADR)

7. Quality checklist

☐ Two options scored with weights
☐ Modular monolith unless justified
☐ ERD normalised and matches I/O spec
☐ Auth, session and RBAC design written
☐ Secrets management defined
☐ Backups and RPO/RTO defined
☐ Cost of each option estimated

8. Approval criteria

G3 signed by Tech Lead and Business Owner. Any later stack change requires an ADR and a Change Request.

9. Official source references (verify and record access date)

- Next.js docs https://nextjs.org/docs
- React docs https://react.dev/
- PostgreSQL docs https://www.postgresql.org/docs/

- OWASP https://owasp.org/
- AWS docs https://docs.aws.amazon.com/
- Vercel docs https://vercel.com/docs

10. Common mistakes & prevention

Common mistake	Prevention
Choosing by hype	Use the weighted scorecard
Premature microservices	Modular monolith with clear module boundaries
No threat model	Mandatory section; reviewed at G3

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4 UI/UX & Prototype

G4 – Design Approval

Finds usability problems when they cost hours, not weeks.

1. Detailed instructions

1. Create sitemap and user-flow diagrams from the PRD journeys.
2. Low-fidelity wireframes → review → high-fidelity designs in Figma.
3. Define design system: type scale, colour tokens (with WCAG AA contrast), spacing, components, states (empty, loading, error).
4. Design responsive layouts (mobile first; 360, 768, 1280 px).
5. Build a clickable prototype and run a usability review with ≥5 target users; log findings and fixes.
6. Prepare a showcase dataset using synthetic or authorised data only.

In practice – precise targets and artefacts

- **Breakpoints:** 360 / 768 / 1280 px. Touch targets ≥ 44×44 px (Apple HIG guidance) / 48 dp (Material).
- **Tokens:** colour, type scale, spacing (4/8 px grid), radius, elevation; text contrast ≥ 4.5:1 (WCAG 2.2 AA).
- **Usability test:** 5 users × 5 tasks; record task success, time, errors, quotes. Fix every task failed by ≥2 users before G4.
- **Performance budget:** JS ≤ 200 KB gzipped on first load for marketing pages; no autoplay animation.

2. Employee responsibilities

Role	Responsibility
UI/UX Designer	Designs, runs usability sessions
Product Owner	Approves flows
Frontend Developer	Reviews feasibility and component API
Customer	Approves the prototype

3. Required tools

Figma, Maze/Loom for tests, axe/Lighthouse for accessibility.

4. Estimated time

2–4 weeks (≈ 113 h)

5. Estimated cost

Labour ≈ ₹79,333 (designer hours).

6. Expected deliverables

- Sitemap + flows
- Wireframes
- Hi-fi designs
- Design system / component library spec
- Clickable prototype
- Usability report

7. Quality checklist

☐ Every PRD journey is covered
☐ WCAG 2.2 AA contrast for text
☐ Keyboard and screen-reader basics considered
☐ All states designed
☐ No decorative animation that hurts performance
☐ Customer sign-off recorded

8. Approval criteria

G4 signed by Product Owner and Customer; changes after G4 follow change control.

9. Official source references (verify and record access date)

- WCAG 2.2 https://www.w3.org/TR/WCAG22/
- web.dev performance https://web.dev/

10. Common mistakes & prevention

Common mistake	Prevention
Skipping error/empty states	Design-system checklist includes all states
Using real customer data in demos	Synthetic data only, unless authorised in writing

5 VS Code, GitHub & Project Initialisation

G5 – Repo Readiness

Every project starts from the same secure, automated baseline.

1. Detailed instructions

1. Install standard tools: VS Code (with approved extensions), Git, approved Node/Python runtime and package manager, API client, DB client, browser dev tools, test runners.
2. Create a PRIVATE GitHub repository under the company organisation; add description and README.
3. Create the standard structure (apps/web, apps/api, packages/ui, packages/shared, docs/*, tests/{unit,integration,e2e}, infra, .github/{workflows,ISSUE_TEMPLATE}, .env.example, .gitignore, README.md, CONTRIBUTING.md).
4. Protect main: require PR, ≥1 approving review, passing status checks, no force-push, linear history.
5. Branches: feature/-, fix/-, hotfix/-. Commits: Conventional Commits (feat:, fix:, docs:, chore:).
6. CI: lint, type-check, unit tests, build, dependency audit, secret scanning, CodeQL/static analysis. Enable Dependabot.
7. Create three environments (local, staging/preview, production) with separate databases and secrets.
8. Tag releases with semantic versions and publish release notes.

In practice – precise targets and artefacts

- **Branch protection:** require PR, ≥1 approval, up-to-date branch, passing checks (lint, types, unit, build, audit, secret scan, code scanning); dismiss stale approvals; no force-push.
- **Environment matrix (names only, never values):** DATABASE_URL, AUTH_SECRET, SMTP/Email key, PAYMENT_KEY, STORAGE_KEY, ERROR_TRACKER_DSN – one set per environment.
- **Release tagging:** semantic versions vMAJOR.MINOR.PATCH; release notes list features, fixes, migrations, rollback note.
- **Onboarding target:** new developer runs the app locally in ≤ 30 minutes from the README.

2. Employee responsibilities

Role	Responsibility
Tech Lead	Owns repo standards and branch protection
DevOps Engineer	CI/CD and environments
All developers	Follow CONTRIBUTING.md

3. Required tools

VS Code, Git, GitHub (Actions, Dependabot, code scanning, secret scanning), Postman/Bruno, DBeaver/pgAdmin, Playwright/Jest/Vitest/Pytest.

4. Estimated time

2–3 days

5. Estimated cost

Labour ≈ 16 h DevOps; GitHub plan cost is a shared company cost (allocate by time share).

6. Expected deliverables

- Configured repository
- CI pipeline green on an empty PR
- CONTRIBUTING.md, PR and issue templates
- Environment matrix (no secrets in the document)

7. Quality checklist

☐ Repo is private
☐ main is protected
☐ Secrets only in provider secret stores
☐ CI blocks merges on failure
☐ Dependabot + secret scanning on
☐ Staging and production are fully separated

8. Approval criteria

G5 signed by Tech Lead after a dry-run PR passes all checks.

9. Official source references (verify and record access date)

- VS Code docs https://code.visualstudio.com/docs
- GitHub docs https://docs.github.com/
- GitHub protected branches https://docs.github.com/en/repositories/configuring-branches-and-merges-in-your-repository/managing-protected-branches
- Conventional Commits https://www.conventionalcommits.org/

10. Common mistakes & prevention

Common mistake	Prevention
Committing .env or keys	Pre-commit secret scan + push protection
Sharing one database across environments	Separate DB and credentials per environment

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6 Application Development

G6 – Release Candidate

Builds in small testable increments with weekly demos, never a big-bang at the end.

1. Detailed instructions

1. Order of work: database → schema → migrations → backend structure → authentication → authorisation/roles → APIs → frontend structure → UI components → business logic → integration → file storage → email/notifications → reports/exports → payments (if required) → admin dashboard → user dashboard → error handling → logging → documentation.
2. Every module carries: requirement reference, owner, due date, acceptance criteria, test cases, code review, documentation.
3. Definition of Done: acceptance criteria pass; unit + integration tests added; code reviewed by someone else; no new critical lint/security findings; docs updated; demoed.
4. Hold a working demo every week to the Product Owner using the staging environment.
5. Track hours against estimate in the dashboard; flag >20% variance immediately.
6. Freeze features; produce the Release Candidate; run the QA and security gates.

In practice – precise targets and artefacts

- **Sprint rhythm:** 1-week increments; Monday plan, Friday demo; demo only on staging with seeded synthetic data.
- **Definition of Done checklist:** criteria pass · tests added · reviewed · no critical/high scanner findings · migration reversible · docs updated · demo accepted.
- **Estimate tracking:** variance = (actual – expected) ÷ expected; >20% on any module triggers a scope or estimate review the same week.
- **Quality gates:** unit coverage ≥ 80% on business logic; zero open Critical/High defects at G6.

2. Employee responsibilities

Role	Responsibility
Developers	Build and test modules
Tech Lead	Architecture compliance, reviews
QA Engineer	Test plans, regression, UAT support
Product Owner	Accepts or rejects each module

3. Required tools

VS Code, GitHub, ORM + migrations, test frameworks, Playwright, Postman, Sentry-type error tracker.

4. Estimated time

8–14 weeks for the sample product

5. Estimated cost

Labour: frontend + backend + auth + integrations + docs ≈ ₹5,91,733 expected.

6. Expected deliverables

- Working software on staging
- Weekly demo notes
- Test reports
- API documentation
- Release notes

7. Quality checklist

☐ No work without a requirement reference
☐ ≥ agreed unit-test coverage on business logic
☐ All endpoints authorised by role
☐ Input validated server-side
☐ Errors logged without leaking PII
☐ Migrations reversible
☐ Docs match behaviour

8. Approval criteria

G6: Product Owner accepts all in-scope modules; Tech Lead certifies no open critical/high defects.

9. Official source references (verify and record access date)

- OWASP Cheat Sheets https://cheatsheetseries.owasp.org/
- Node.js docs https://nodejs.org/docs/
- PostgreSQL docs https://www.postgresql.org/docs/

10. Common mistakes & prevention

Common mistake	Prevention
Client-only validation	Duplicate every rule on the server
Merging without review	Branch protection requires ≥1 review
Hidden scope creep	Reject work without a requirement/CR ID

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7 Production Cost Calculation (mandatory)

G7 – Cost Model Approval

Prevents under-pricing by costing development, launch AND operation. Never quote from domain + hosting alone.

1. Detailed instructions

1. Labour: hours × fully loaded hourly cost per role, three scenarios (optimistic, expected via PERT = (O + 4M + P) / 6, pessimistic). Include founder time.
2. One-time infrastructure/setup: domain, DNS, hosting, DB, email, SSL, storage, monitoring, payment gateway, backups, pipeline, legal, integrations. Mark each mandatory or optional.
3. Monthly cost: domain, hosting, database, storage, CDN, email, SMS/WhatsApp, auth, APIs, AI, monitoring, backups, security, payment fees, maintenance, support, misc.
4. Pull every unit price from the provider's current official pricing page; store URL + access date. If unavailable mark ASSUMPTION. Never invent a price.
5. Capacity: model 10, 50, 100, 500, 1,000, 5,000, 10,000 users via MAU, DAU, peak concurrency, requests, queries, storage, bandwidth, emails, background jobs. Use low/medium/high scenarios if no data. Find the real bottleneck; do not multiply linearly.
6. Cost three architectures: Lean MVP, Professional, Scalable. Report setup, monthly infra, maintenance, annual, capacity, limits, risks, upgrade triggers.
7. Apply the formulas in section 'Financial Model' and allocate shared costs once (usage, reserved capacity or time share).

In practice – precise targets and artefacts

- **Order of calculation:** labour → one-time setup → monthly run cost → capacity → three models → first-year total → shared-cost allocation.
- **Price evidence rule:** store provider · plan · unit · price · currency · region · access date · URL. Any figure older than 90 days is re-verified before G7.
- **Bottleneck check:** compare peak req/s, DB connections, storage and bandwidth with plan limits; cost the first limit reached.
- **Sensitivity:** re-run with usage ×0.5 and ×2; any plan whose gross margin drops below 50% is re-priced or limited.

2. Employee responsibilities

Role	Responsibility
Finance Analyst / Founder	Owns the model
Tech Lead	Usage and capacity assumptions
DevOps Engineer	Provider pricing and architecture costs

3. Required tools

Cost workbook (this document's model), provider pricing calculators (AWS, GCP, Azure, Vercel, Cloudflare), dashboard cost calculator.

4. Estimated time

3–5 days

5. Estimated cost

Labour ≈ 40 h; included in company overhead.

6. Expected deliverables

- Production Cost & SaaS Pricing Financial Model
- Assumption register with source URLs
- Three-model comparison

7. Quality checklist

☐ Every price has URL + date or is tagged ASSUMPTION
☐ One-time ≠ monthly
☐ Free tiers not treated as permanent
☐ Shared costs counted once
☐ Maintenance and support labour included
☐ Optimistic/expected/pessimistic shown
☐ Capacity bottleneck identified

8. Approval criteria

G7 signed by Business Owner and Finance. Model is re-run on any scope change or provider price change.

9. Official source references (verify and record access date)

- AWS Pricing Calculator https://calculator.aws/
- Vercel pricing https://vercel.com/pricing
- Cloudflare plans https://www.cloudflare.com/plans/
- Google Cloud pricing calculator https://cloud.google.com/products/calculator
- Azure pricing calculator https://azure.microsoft.com/pricing/calculator/

10. Common mistakes & prevention

Common mistake	Prevention
Quoting only hosting	Use full formula incl. labour and support
Invented prices	ASSUMPTION tag or cite URL
Linear scaling	Use bottleneck-driven tiers

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8 SaaS Pricing & Profitability

G8 – Pricing Approval

Sets a price that covers full cost, CAC and churn with a target margin.

1. Detailed instructions

1. Cost-based: (cost to serve + fixed share + target margin). Competitor-based: positioning within the observed range. Value-based: share of customer's quantified saving. Choose the price inside the intersection.
2. Compute CAC, MRR, ARR, gross margin (state which costs sit in cost of revenue), contribution margin, break-even customers, operating profit; show tax separately.
3. Define plans: Free Trial (if suitable), Starter, Professional, Business, Enterprise — price (monthly/annual), users, storage, API limits, features, support, overage, upgrade rule.
4. Include payment-gateway fees, refunds, discounts, GST, failed payments, churn.
5. Never promise 'unlimited' without a cost and abuse-control rule (fair-use limit, rate limiting).
6. Recommend setup fee, subscription fees, minimum viable price, target gross margin, maximum CAC, break-even customers, monthly revenue target, renewal strategy.

In practice – precise targets and artefacts

- **Margin gate:** LTV:CAC ≥ 3, CAC payback < 12 months, gross margin ≥ 60% at the planned scale.
- **Price ladder logic:** each plan roughly 2.5× the previous; limits chosen so ≥80% of target customers fit the plan they are sold.
- **Leakage:** subtract refunds, failed payments, gateway fee, discounts before contribution; GST is a pass-through, not revenue.
- **Store channel:** if the product is also sold inside iOS/Android apps, use the 'App store fees' section; commissions replace the gateway fee on those sales.

2. Employee responsibilities

Role	Responsibility
Business Owner	Decides price
Finance Analyst	Maintains the model
Sales Lead	Validates price with prospects

3. Required tools

Pricing workbook / dashboard SaaS Pricing Calculator.

4. Estimated time

1 week

5. Estimated cost

Labour ≈ 30 h.

6. Expected deliverables

- Pricing Model
- Plan matrix
- Profitability table by customer count
- Commercial recommendation

7. Quality checklist

☐ Three pricing approaches compared
☐ Margin target stated
☐ LTV/CAC ≥ 3 and payback < 12 months
☐ Break-even customer count explained
☐ Fair-use and overage defined

8. Approval criteria

G8 signed by Business Owner. Prices are published only after G8.

9. Official source references (verify and record access date)

- CBIC – GST https://cbic-gst.gov.in/
- Razorpay pricing (example gateway) https://razorpay.com/pricing/
- Stripe pricing (example gateway) https://stripe.com/pricing

10. Common mistakes & prevention

Common mistake	Prevention
Pricing off competitors only	Triangulate with cost and value
Ignoring churn and refunds	Apply leakage factors

Common mistake	Prevention
Calling revenue profit	Show operating profit and tax separately

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9 Domain, DNS & Production Deployment

G9 – Release Approval

Takes the product from repository to a safe, monitored, rollback-able live service.

1. Detailed instructions

1. Domain: generate options, check availability at reputable registrars (never claim availability without checking), compare first-year vs renewal price, check trademark conflicts, register in the COMPANY's name, enable MFA and auto-renew/renewal lock, record ownership and recovery procedure.
2. DNS: copy exact records supplied by the hosting and email providers (A/AAAA/CNAME/MX/TXT/SPF/DKIM/DMARC/CAA). Keep existing email records. Never guess a value.
3. Deploy: connect repo → build settings → production env vars → production DB → storage → domain → DNS → verify propagation → TLS → HTTPS redirect → migrations → backups → monitoring → error tracking → alerts → smoke tests → critical workflow tests → rollback test → release approval.
4. Record every value in the Domain/DNS Report and Deployment Report.

In practice – precise targets and artefacts

- **Cut-over order:** lower DNS TTL (e.g. 300 s) 24 h before → deploy → verify on temporary URL → switch records → verify TLS and redirects → raise TTL.
- **Email auth targets:** SPF pass, DKIM pass, DMARC policy published (start p=none, review reports, move to quarantine/reject).
- **Smoke tests (≤15 min):** sign-up, login, core create/read/update, payment (test mode then one live ₹ transaction refunded), email delivered, export, logout.
- **Rollback proof:** previous artifact redeployed and DB migration reversed on staging within 15 minutes, evidence attached.

2. Employee responsibilities

Role	Responsibility
DevOps Engineer	Executes SOP
Tech Lead	Verifies and approves
Business Owner	Registrar account owner, release approval

3. Required tools

Registrar console, DNS provider, hosting platform, dig/nslookup, SSL Labs, uptime monitor, error tracker.

4. Estimated time

2–4 days

5. Estimated cost

Domain + hosting per cost model; labour ≈ 36 h DevOps in sample.

6. Expected deliverables

- Domain/DNS Report
- Deployment Report
- Smoke-test evidence
- Rollback test evidence

7. Quality checklist

☐ MFA on registrar, DNS and hosting accounts
☐ Auto-renew on; renewal reminders in shared calendar
☐ SPF, DKIM, DMARC verified
☐ HTTPS-only with valid certificate
☐ Backup restore tested
☐ Alerts reach on-call
☐ Rollback executed on staging

8. Approval criteria

G9 signed by Tech Lead and Business Owner after smoke tests pass.

9. Official source references (verify and record access date)

- ICANN https://www.icann.org/
- Cloudflare DNS docs https://developers.cloudflare.com/dns/
- Vercel domains https://vercel.com/docs/domains
- DMARC.org https://dmarc.org/

10. Common mistakes & prevention

Common mistake	Prevention
Overwriting MX records	Export existing zone before edits
Registering domain in an employee's name	Company account only
No restore test	Restore backups to staging quarterly

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10 Customer Showcase, Sales & Delivery

G10 – Commercial Approval

Turns a working product into signed, paying, correctly-onboarded customers.

1. Detailed instructions

1. Identify and qualify leads against the ICP; log in the CRM; schedule discovery calls.
2. Showcase script: introduction → customer problem → current workflow → solution → live demo → benefits → limitations → pricing → pilot proposal → next steps.
3. Demo only completed features, with synthetic or authorised data. Get written consent before using any logo, name, screenshot or testimonial.
4. Proposal → SOW → timeline → agreement → SaaS subscription terms → onboarding → user and admin manuals → training → acceptance form → support details.
5. Contract must define ownership of source code, reusable components, third-party libraries, customer data and IP.

In practice – precise targets and artefacts

- **Funnel metrics:** lead → qualified → demo → pilot → paid; record conversion and days in stage to calibrate CAC and sales cycle.
- **Demo rule:** 25 minutes demo, 5 minutes limitations, 10 minutes pricing/next steps; leave a one-page summary.
- **Pilot terms:** 30–60 days, 2–3 named success metrics, a conversion price agreed before start.
- **Contract must state:** scope, fees, term/renewal, data ownership and export, IP, liability cap, termination, support hours, governing law.

2. Employee responsibilities

Role	Responsibility
Sales Lead	Pipeline and proposals
Product Owner	Demo and scoping
Support Lead	Onboarding and training
Legal advisor	Contract review

3. Required tools

CRM / dashboard pipeline, demo environment, e-signature, proposal templates.

4. Estimated time

2–8 weeks per sale (depends on sales cycle)

5. Estimated cost

Included in CAC.

6. Expected deliverables

- Demo environment + script
- Proposal and SOW
- Agreement and subscription terms
- Customer acceptance form
- Handover pack

7. Quality checklist

☐ No unfinished feature shown as complete
☐ Written consent for logos
☐ IP clauses reviewed
☐ Acceptance criteria tied to PRD
☐ Support hours documented

8. Approval criteria

G10: discount or special terms beyond policy need Business Owner approval.

9. Official source references (verify and record access date)

- MeitY https://www.meity.gov.in/
- CBIC GST https://cbic-gst.gov.in/

10. Common mistakes & prevention

Common mistake	Prevention
Promising roadmap items	State limitations explicitly
Verbal scope agreements	Everything in the SOW

11 Maintenance & Post-Launch Operations

G11 – Operational Readiness

Recurring revenue only survives if the service stays up, secure and supported.

1. Detailed instructions

1. Monitoring: uptime, application errors, database health, queue depth, certificate and domain expiry.
2. Monthly routine: security patching, dependency updates, backup restore verification, cost review, usage analytics, subscription/renewal review.
3. Incident levels: Critical (service down or confirmed serious security incident), High (major function unavailable), Medium (limited impact), Low (minor/cosmetic). Each has owner, escalation, communication and recovery steps (see Incident Matrix).
4. Maintenance agreement: service hours, response targets, exclusions, price. Only promise what the on-call rota can actually deliver.

In practice – precise targets and artefacts

- **Monthly calendar:** week 1 patches & dependency updates; week 2 backup restore drill; week 3 cost & usage review; week 4 customer report.
- **Alerts:** uptime check every 1–5 min from two locations; error-rate and latency thresholds; certificate/domain expiry ≥ 30 days notice.
- **Post-incident review:** timeline, root cause, customer impact, fix, prevention; blameless; completed within 5 working days of Critical/High incidents.
- **Support KPIs:** first response time, resolution time, reopen rate, CSAT.

2. Employee responsibilities

Role	Responsibility
Support Lead	Tickets and customer comms
DevOps Engineer	Monitoring, patching, recovery
Tech Lead	Incident commander for Critical/High

3. Required tools

Uptime monitor, error tracker, log platform, ticketing, status page.

4. Estimated time

Ongoing; budget 20 maintenance hours/month in Professional model

5. Estimated cost

≈ ₹17,000/month maintenance labour + support (see model).

6. Expected deliverables

- Maintenance Agreement
- Runbooks
- Monthly Maintenance Report
- Post-incident reviews

7. Quality checklist

☐ On-call rota exists
☐ Backups verified monthly
☐ Patches applied within agreed window
☐ Customer comms templates ready

8. Approval criteria

G11 signed by Tech Lead; SLA wording reviewed by Business Owner.

9. Official source references (verify and record access date)

- Google SRE book https://sre.google/sre-book/table-of-contents/
- NIST SSDF https://csrc.nist.gov/pubs/sp/800/218/final

10. Common mistakes & prevention

Common mistake	Prevention
Offering 99.99% without redundancy	SLA must match architecture and staffing
Never testing restores	Scheduled restore drill

12 Employee Management & Company SOP

Continuous

Makes every project run the same way regardless of who works on it.

1. Detailed instructions

1. Every project record holds: project ID, product owner, tech lead, developers, QA owner, customer contact, budget, timeline, approved scope, cost estimate, sales price, expected margin, current stage, risk register.
2. Weekly review: each employee reports completed, pending, blockers, time spent, quality issues, upcoming deliverables.
3. All 20 standard templates are used unchanged unless the Tech Lead approves a variant.

In practice – precise targets and artefacts

- **Weekly review agenda (30 min):** gate status · variance · blockers · risks · approvals due.
- **Timesheet rule:** hours logged daily against project ID and stage; unlogged time is not reportable.
- **Risk score:** likelihood (1–5) × impact (1–5); ≥15 needs a named owner and mitigation date.

2. Employee responsibilities

Role	Responsibility
Managing Director	Final authority
Project manager / Product Owner	Weekly review
Employees	Timely accurate reporting

3. Required tools

Internal dashboard, timesheets, shared drive.

4. Estimated time

30 min weekly review per project

5. Estimated cost

Overhead.

6. Expected deliverables

- Project record
- Weekly status report
- Risk register

7. Quality checklist

☐ Every active project has all 14 fields
☐ Timesheets filled daily
☐ Risks reviewed weekly

8. Approval criteria

Reviewed weekly by management.

9. Official source references (verify and record access date)

- Atlassian agile guides https://www.atlassian.com/agile

10. Common mistakes & prevention

Common mistake	Prevention
Status by chat only	Dashboard is the source of truth

13 Internal Management Dashboard

G13 – Dashboard Release

Gives management live visibility of pipeline, delivery, cost, revenue and approvals.

1. Detailed instructions

1. Build the portal as a standard SaaS project using this same process (it is Project 0).
2. Implement the 20 modules and role-based access (see Dashboard Specification).

In practice – precise targets and artefacts

- **Build order:** auth/roles → projects & stage tracker → approvals → time/cost → pipeline & subscriptions → analytics.
- **Audit trail:** every approval stores user, timestamp, decision, comment, document version.
- **KPI integrity:** each KPI has an owner, definition, source table and refresh time on its tooltip.

2. Employee responsibilities

Role	Responsibility
Tech Lead	Architecture
Developers	Modules
Managing Director	Product owner

3. Required tools

Next.js/React + TypeScript + PostgreSQL (suggested, subject to G3).

4. Estimated time

8–12 weeks

5. Estimated cost

Internal project; track like any other.

6. Expected deliverables

- Dashboard v1 with all management KPIs

7. Quality checklist

☐ Role-based access enforced server-side
☐ Approvals recorded with timestamp and user
☐ KPIs reproducible from raw data

8. Approval criteria

G13 by Managing Director.

9. Official source references (verify and record access date)

- OWASP ASVS https://asvs.owasp.org/

10. Common mistakes & prevention

Common mistake	Prevention
Vanity metrics	Every KPI has a definition and owner

14 Source Documentation Discipline

Continuous

Every technical or financial claim is traceable.

1. Detailed instructions

1. Prefer official docs, pricing pages, registrar, DNS, cloud calculators, gateways, security standards, government tax/legal sources.
2. Store each source as: title, URL, publisher, access date, what it supports.
3. Re-verify pricing and regulation at every G7/G8 and before every contract.

In practice – precise targets and artefacts

- **Source register columns:** claim · source title · publisher · URL · access date · verified by.
- **Priority:** official docs → official pricing → standards bodies → government → reputable secondary (as leads only).
- **Staleness rule:** prices > 90 days, regulations > 180 days old must be re-verified before use in a quote or contract.

2. Employee responsibilities

Role	Responsibility
Everyone	Cite sources
Tech Lead	Spot-checks

3. Required tools

Shared source register.

4. Estimated time

Continuous

5. Estimated cost

Overhead.

6. Expected deliverables

- Source register per project

7. Quality checklist

☐ No uncited price
☐ No uncited legal claim
☐ Access date present

8. Approval criteria

Checked at each gate.

9. Official source references (verify and record access date)

- See Source Index appendix #sources

10. Common mistakes & prevention

Common mistake	Prevention
Using blog posts for prices	Use the official page; blogs only as leads

Financial model – cost, pricing, profitability (computed)

This section implements Phases 7 and 8 for the sample product. **Every unit price is an ASSUMPTION** until replaced by a cited official price.

A. One-time development cost – labour

`Development Labour Cost = Estimated Hours × Fully Loaded Hourly Cost`; Expected hours use PERT = (O + 4M + P) ÷ 6. Founder time is included.

Role	Fully loaded ₹/hour (ASSUMPTION)
Product Owner / Founder	₹1,200
Tech Lead	₹1,500
Full-stack Developer	₹800
UI/UX Designer	₹700
QA Engineer	₹500
DevOps Engineer	₹900

Fully loaded = salary or founder-equivalent pay + employer taxes/benefits + equipment + software seats + office/overheads, divided by productive (not paid) hours. Replace with your actuals.

Activity	Role	O h	Expected h (PERT)	P h	₹/h	Optimistic	Expected	Pessimistic
Market research	Product Owner / Founder	30	41.7	60	₹1,200	₹36,000	₹50,000	₹72,000
Requirement analysis & PRD	Product Owner / Founder	40	61.7	90	₹1,200	₹48,000	₹74,000	₹1,08,000
Requirement analysis & PRD	Tech Lead	10	16.3	24	₹1,500	₹15,000	₹24,500	₹36,000
UI/UX design	UI/UX Designer	80	113.3	160	₹700	₹56,000	₹79,333	₹1,12,000
Frontend development	Full-stack Developer	200	286.7	400	₹800	₹1,60,000	₹2,29,333	₹3,20,000
Backend development	Full-stack Developer	220	310.0	440	₹800	₹1,76,000	₹2,48,000	₹3,52,000
Database & API design	Tech Lead	40	61.7	90	₹1,500	₹60,000	₹92,500	₹1,35,000
Authentication & RBAC	Full-stack Developer	30	41.7	60	₹800	₹24,000	₹33,333	₹48,000
Integrations (payments, email, SMS)	Full-stack Developer	40	63.3	100	₹800	₹32,000	₹50,667	₹80,000
Testing	QA Engineer	100	146.7	220	₹500	₹50,000	₹73,333	₹1,10,000
Security review	Tech Lead	16	25.3	40	₹1,500	₹24,000	₹38,000	₹60,000
Deployment & DevOps	DevOps Engineer	24	38.0	60	₹900	₹21,600	₹34,200	₹54,000
Documentation	Full-stack Developer	24	38.0	60	₹800	₹19,200	₹30,400	₹48,000
Customer onboarding	Product Owner / Founder	16	25.3	40	₹1,200	₹19,200	₹30,400	₹48,000
Total		**870**	**1270**	**1844**		**₹7,41,000**	**₹10,88,000**	**₹15,83,000**

B. One-time infrastructure & setup cost

Item	Amount (ASSUMPTION)	Type	Note
Domain registration (year 1)	₹1,000	Mandatory	Assumption; check registrar price page for the exact TLD
Legal: T&Cs, Privacy Policy, DPA review	₹25,000	Mandatory	Assumption; obtain lawyer quote
Hosting / DB / storage / SSL / DNS setup	₹0	Mandatory	Labour already in DevOps hours; provider setup fees assumed nil
Payment-gateway onboarding	₹0	Mandatory	Assumed nil setup fee; verify on gateway pricing page
Trademark search & filing	₹8,000	Optional	Optional; assumption
External penetration test	₹40,000	Optional	Optional but recommended before enterprise sales; assumption
Total mandatory	**₹26,000**		
Total optional	**₹48,000**		

C/D. Usage assumptions and capacity scenarios

Assumption	Value
Monthly active users (of registered)	80%
Daily active users (of registered)	50%
Requests per DAU per day	150
Share of daily requests in busiest hour	15%
Burst factor on peak hour	3×
Peak concurrency	10% of DAU
DB queries per request	4
Storage per user	50 MB
Bandwidth per DAU per day	3 MB
Emails per user per month	40
SMS/WhatsApp per user per month	60

Low/High scenarios: halve or double the request, storage and messaging values; re-run the model.

Users	MAU	DAU	Requests/day	Peak req/s (3× burst)	Peak concurrent	DB queries/day	Storage GB	Bandwidth GB/mo	Emails/mo	SMS+WA msgs/mo
10	8	5	750	0.1	0	3,000	0.5	0	400	600
50	40	25	3,750	0.5	2	15,000	2.4	2	2,000	3,000
100	80	50	7,500	0.9	5	30,000	4.9	4	4,000	6,000
500	400	250	37,500	4.7	25	150,000	24.4	22	20,000	30,000
1,000	800	500	75,000	9.4	50	300,000	48.8	44	40,000	60,000
5,000	4,000	2,500	375,000	46.9	250	1,500,000	244.1	220	200,000	300,000
10,000	8,000	5,000	750,000	93.8	500	3,000,000	488.3	439	400,000	600,000

Unit-rate assumptions

Unit rate (ALL ASSUMPTIONS)	Value
Object storage	₹4 / GB-month
Bandwidth / CDN egress	₹8 / GB
Transactional email	₹80 / 1,000 emails
SMS / WhatsApp	₹0.5 / message
Maintenance labour	₹850 / hour
Support labour	₹600 / hour; 0.5 h per tenant per month
Lean MVP: base platform	₹2,500 / month incl. app host, managed DB, backups, monitoring
Lean MVP: extra compute unit	₹1,500 per unit; one unit sustains 10 req/s
Professional: base platform	₹9,000 / month incl. app host, managed DB, backups, monitoring
Professional: extra compute unit	₹4,000 per unit; one unit sustains 40 req/s
Scalable: base platform	₹35,000 / month incl. app host, managed DB, backups, monitoring
Scalable: extra compute unit	₹8,000 per unit; one unit sustains 100 req/s

Replace before use
Every rate above is a planning placeholder to make the model computable. Before Gate G7, replace each with the provider's current official price (URL + access date) using the Cost Estimation template. Do not quote customers from these numbers.

E. Three production models – monthly operating cost by user count

Cost is bottleneck-driven: compute is stepped by peak requests/second (units needed = ⌈peak ÷ unit capacity⌉), storage/bandwidth/messages scale with usage, maintenance is a fixed labour block, support scales with tenants (users ÷ 7.3 per clinic). Not a straight line.

Lean MVP

Users	Compute units	Platform + compute	Usage (storage, BW, email, msgs)	Maintenance labour	Support labour	Monthly operating	Annual	Status
10	1	₹2,500	₹337	₹10,200	₹411	**₹13,448**	₹1,61,381	OK
50	1	₹2,500	₹1,687	₹10,200	₹2,055	**₹16,442**	₹1,97,306	OK
100	1	₹2,500	₹3,375	₹10,200	₹4,110	**₹20,184**	₹2,42,211	OK
500	1	₹2,500	₹16,873	₹10,200	₹20,548	**₹50,121**	₹6,01,457	OK
1,000	1	₹2,500	₹33,747	₹10,200	₹41,096	**₹87,543**	₹10,50,513	OK
5,000	5	₹8,500	₹1,68,734	₹10,200	₹2,05,479	**₹3,92,914**	₹47,14,966	Outgrown – upgrade
10,000	10	₹16,000	₹3,37,469	₹10,200	₹4,10,959	**₹7,74,628**	₹92,95,532	Outgrown – upgrade

Professional

Users	Compute units	Platform + compute	Usage (storage, BW, email, msgs)	Maintenance labour	Support labour	Monthly operating	Annual	Status
10	1	₹9,000	₹337	₹17,000	₹411	**₹26,748**	₹3,20,981	OK
50	1	₹9,000	₹1,687	₹17,000	₹2,055	**₹29,742**	₹3,56,906	OK
100	1	₹9,000	₹3,375	₹17,000	₹4,110	**₹33,484**	₹4,01,811	OK
500	1	₹9,000	₹16,873	₹17,000	₹20,548	**₹63,421**	₹7,61,057	OK
1,000	1	₹9,000	₹33,747	₹17,000	₹41,096	**₹1,00,843**	₹12,10,113	OK
5,000	2	₹13,000	₹1,68,734	₹17,000	₹2,05,479	**₹4,04,214**	₹48,50,566	OK
10,000	3	₹17,000	₹3,37,469	₹17,000	₹4,10,959	**₹7,82,428**	₹93,89,132	OK

Scalable

Users	Compute units	Platform + compute	Usage (storage, BW, email, msgs)	Maintenance labour	Support labour	Monthly operating	Annual	Status
10	1	₹35,000	₹337	₹34,000	₹411	**₹69,748**	₹8,36,981	OK
50	1	₹35,000	₹1,687	₹34,000	₹2,055	**₹72,742**	₹8,72,906	OK
100	1	₹35,000	₹3,375	₹34,000	₹4,110	**₹76,484**	₹9,17,811	OK
500	1	₹35,000	₹16,873	₹34,000	₹20,548	**₹1,06,421**	₹12,77,057	OK
1,000	1	₹35,000	₹33,747	₹34,000	₹41,096	**₹1,43,843**	₹17,26,113	OK
5,000	1	₹35,000	₹1,68,734	₹34,000	₹2,05,479	**₹4,43,214**	₹53,18,566	OK
10,000	1	₹35,000	₹3,37,469	₹34,000	₹4,10,959	**₹8,17,428**	₹98,09,132	OK

Model comparison

Model	Typical build	Estimated capacity	Limitations / risks	Upgrade triggers
Lean / Low-cost MVP	Managed PaaS free/low tier + small managed Postgres, daily backup, basic uptime + error tracking	≈ up to 1,000 users (peak < 10 req/s)	Free tiers can change or be withdrawn; shared resources; limited backups; single region; no redundancy	Paying customers > 25, any SLA promise, peak load > 70% of unit, DB > 70% of storage cap
Professional	Paid PaaS/VM, managed Postgres with point-in-time recovery, staged environments, monitoring, security scanning	≈ up to 5,000 users with extra compute units	Single region; manual failover; cost grows stepwise	Enterprise customers, uptime > 99.5% contractually, p95 latency breaches, DB CPU > 60%
Scalable	Multi-AZ app + DB, autoscaling, CDN/WAF, centralised logs, DR in second region, 24×7 alerting	10,000+ users	Highest fixed cost and operations load; needs on-call staff	Revenue/MRR > monthly cost × 3 and enterprise commitments

Model (at 1,000 users)	One-time setup (non-labour)	Monthly infrastructure	Monthly maintenance	Monthly operating (incl. support)	Annual recurring
Lean MVP	₹26,000	₹36,247	₹10,200	₹87,543	₹10,50,513
Professional	₹26,000	₹42,747	₹17,000	₹1,00,843	₹12,10,113
Scalable	₹26,000	₹68,747	₹34,000	₹1,43,843	₹17,26,113

F. Complete cost formula

```
Total One-Time Cost   = Research + Design + Development + Testing + Setup + Deployment +
Legal/Compliance
Monthly Operating Cost= Hosting + Database + Storage + Bandwidth + APIs + Email + Monitoring + Backups +
Maintenance + Support + Other
Annual Operating Cost = Monthly Operating Cost × 12
Total First-Year Cost = Total One-Time Cost + Annual Operating Cost
```

Item	Amount
Total one-time cost (labour expected + mandatory setup)	₹11,14,000
Monthly operating cost @100 users, Professional	₹33,484
Annual operating cost = monthly × 12	₹4,01,811
Total first-year cost = one-time + annual operating	**₹15,15,811**
Range using pessimistic labour	₹20,10,811
Range using optimistic labour	₹11,68,811

Shared vs product-specific: Shared company items (GitHub org, design tool seats, password manager, accounting software) are totalled once and allocated to products by employee-time share (example: ₹10,000 shared × 25% = ₹2,500 charged to the product). Product-specific items (its hosting, DB, domain, SMS) are charged 100%. A cost line is never both allocated and charged directly.

Pricing & Profitability (Phase 8)

Pricing approaches

Approach	Method	Result for sample
Cost-based	(Cost to serve + share of fixed cost) ÷ (1 − target margin)	Starter cost to serve ₹401 + fixed share at 100 tenants ₹416 = ₹817 break-even price; with 30% margin ≈ ₹1,167 (before gateway/leakage) – so ₹999 is a thin-margin entry plan, upsell to Professional
Competitor-based	Place inside the observed range from ≥5 competitor pricing pages	*Pending research (no competitor prices invented)*
Value-based	10–20% of customer's quantified saving (e.g. recovered no-show revenue)	If a clinic recovers 5 visits/month × ₹500 = ₹2,500, a ₹999 plan captures ~40%: validate in discovery

Plans

Plan	Monthly (ex-GST)	Annual (10× monthly = 2 months free)	Users	Storage	API calls/mo	Support	Cost to serve + leakage	Contribution / month	Contribution %
Free Trial	₹0 (14 days)	–	2	0.5 GB	2,000	Docs only	₹31 variable (excl. support time)	–	Counted in CAC
Starter	₹999	₹9,990	3	2 GB	10,000	Email, 2 business days	₹470	₹529	53%
Professional	₹2,499	₹24,990	8	10 GB	50,000	Email + chat, 1 business day	₹742	₹1,757	70%
Business	₹5,999	₹59,990	20	50 GB	250,000	Priority email/chat, 4 business hours	₹1,689	₹4,310	72%
Enterprise	Custom (from ≈ ₹15,000)	Custom	Custom	Custom	Custom	Named contact, SLA by contract	Quote per deal	≥ 60% target	Contract

Overage (all plans): extra user ₹150/month; extra 10 GB ₹100/month; extra 10,000 API calls ₹200; SMS/WhatsApp at provider cost + 20%. Upgrade condition: usage > 80% of any limit for 2 consecutive months triggers an upgrade prompt. All prices are ASSUMPTIONS to be validated in discovery.

Fixed monthly cost allocated to the product

Fixed monthly cost (allocated to product)	₹ / month
Professional-model hosting & DB base	₹9,000
Maintenance labour (20 h x ₹850)	₹17,000
Domain (annualised)	₹83
Accounting / legal retainer (allocated)	₹3,000
Shared dev tools (allocated by time share 25% of ₹10,000)	₹2,500
Fixed marketing	₹10,000
Monthly fixed operating cost	**₹41,583**

Key metrics

`CAC = S&M acquisition cost ÷ new paying customers` · `MRR = Σ active subscriptions` · `ARR = MRR × 12` · `Gross margin % = (Revenue − Cost of revenue) ÷ Revenue × 100` · `Contribution = Revenue − variable delivery costs` · `Break-even = Fixed ÷ Contribution per customer (round up)` · `Operating profit = Revenue − all operating expenses`. Cost of revenue here = platform hosting base + maintenance labour + variable delivery + support + gateway fees + refunds/failed-payment leakage; sales/marketing and G&A are operating expenses.

Metric	Value
Blended ARPA (plan mix 50/35/15)	₹2,274 / tenant / month
Contribution per tenant (after gateway fees, refunds, failed payments, variable delivery, support)	₹1,526
Monthly fixed operating cost	₹41,583
Break-even customers = fixed ÷ contribution = ₹41,583 ÷ ₹1,526	**27.25 → round UP to 28 paying tenants**
Churn (ASSUMPTION)	4% / month → lifetime 25 months
LTV = contribution × lifetime	₹38,144
CAC (ASSUMPTION; = sales+marketing spend ÷ new paying customers)	₹6,000
LTV : CAC	6.4 : 1 (target ≥ 3)
CAC payback	3.9 months (target < 12)
Maximum acceptable CAC (LTV ÷ 3)	₹12,715
Revenue leakage assumed	refunds 2% + failed payments 3%
Payment gateway fee (ASSUMPTION; verify gateway page)	2% of collected amount
GST on subscriptions	18% collected from customer, remitted to government – NOT revenue (confirm rate/place-of-supply with CA & CBIC)
Monthly revenue target for 12-month goal	₹2,27,400 MRR (100 tenants)

Profitability by paying-tenant count

Paying tenants	MRR	ARR	Contribution	Operating profit (before tax)	Income tax @25.17% (ASSUMPTION; confirm with CA)	Profit after tax	Gross margin
10	₹22,740	₹2,72,880	₹15,257	-₹26,326	₹0	-₹26,326	-47%
25	₹56,850	₹6,82,200	₹38,144	-₹3,440	₹0	-₹3,440	21%
28	₹63,672	₹7,64,064	₹42,721	₹1,137	₹286	₹851	26%
50	₹1,13,700	₹13,64,400	₹76,287	₹34,704	₹8,735	₹25,969	44%
100	₹2,27,400	₹27,28,800	₹1,52,574	₹1,10,991	₹27,936	₹83,054	56%
250	₹5,68,500	₹68,22,000	₹3,81,435	₹3,39,852	₹85,541	₹2,54,311	63%
500	₹11,37,000	₹1,36,44,000	₹7,62,871	₹7,21,288	₹1,81,548	₹5,39,739	65%

Tax is shown separately: the model applies a flat assumed rate only to positive operating profit and ignores losses carried forward, GST input credit and founder remuneration rules – have a chartered accountant confirm.

Commercial recommendation

Recommendation	Value
Initial setup / onboarding fee	₹5,000 (Starter) · ₹10,000 (Professional) · ₹25,000 (Business) – covers data import and training; recovers CAC
Monthly subscription	₹999 / ₹2,499 / ₹5,999 (ex-GST)
Annual subscription	10× monthly (2 months free); billed upfront – improves cash flow and reduces churn
Minimum viable selling price	Starter ₹999: contribution ₹529 = 53% of price; contribution is still positive at ₹799 but leaves almost no room for CAC, so do not discount below it
Target gross margin	≥ 60% – model reaches 56% at 100 tenants and 63% at 250 tenants; below 60% until roughly 175 tenants
Maximum CAC	₹12,715 (LTV ÷ 3)
Break-even customers	28 paying tenants
Renewal strategy	90/60/30-day renewal reminders; annual plans auto-renew with notice; quarterly business review for Business/Enterprise; price rises ≤ 10% with 60-day notice

App store fees (mobile distribution)

Applies only if the product is also distributed as an iOS or Android app and sells digital goods or subscriptions inside the app. A browser-based B2B SaaS billed through your own gateway has no store commission, but check each store's current rules before deciding the billing channel. Fees below were checked on **2026-11-03**.

Apple – verified from official Apple pages
Fetched directly on 2026-11-03 from developer.apple.com/programs/whats-included and developer.apple.com/app-store/small-business-program.

Item	Official wording / value	Notes
Developer Program fee	"99 USD per membership year, or in local currency where available"	The page gives no INR amount; read the local price at enrolment. Fee waiver exists for eligible nonprofits, accredited educational institutions and government entities.
Standard commission	30% on sale of digital goods and services	Page states 30%.
Small Business Program	15% on paid apps and In-App Purchases	Eligible if proceeds ≤ USD 1 million in the prior calendar year (all associated accounts) and ≤ USD 1 million in the current year; above that the standard rate applies to future sales; can re-qualify the following year. Rate change applies 15 days after the end of the fiscal month of approval.
Qualifying subscriptions	15%	Page lists 15% for qualifying subscriptions.
10% rate	"further reduced commission of 10%" for EU alternative-terms developers in the Small Business Program and for subscriptions after their first year	The sentence is ambiguous about who gets 10%; confirm on the page before modelling it.
Regional variations	Different rates/fees may apply in Brazil, EU, Japan, Netherlands, Russia, South Korea	India is not in that list on the page checked.

Google Play – NOT directly verified
Google's help pages (support.google.com, play.google.com) could not be opened from this build environment (network block). The figures below come from web-search summaries of Google's own pages and **must be confirmed on the linked page before use**. Google has changed its fee model in 2026 and the sources below describe it differently by region and date.

Item	Reported (search summary, 2026-11-03)	Verify at
Developer registration	One-time USD 25 (reported on Google's EEA access terms page)	Play Console Help / Play Console sign-up page
Older structure	15% on the first USD 1M of annual earnings, 30% above; 15% on auto-renewing subscriptions	Service fees

Item	Reported (search summary, 2026-11-03)	Verify at
2026 structure	Fee depends on whether the user's install is 'new' or 'existing' relative to a regional start date; first USD 1M at 10%; auto-renewing subscriptions 10%; other in-app purchases reported at 20% (new) / 25% (existing); billing fee reported at 5% where Google Play Billing is used (US/UK/EEA)	Understanding Google Play's lower service fees; Android Developers Blog
Reported start dates	30 Jun 2026: EEA, UK, US · 30 Sep 2026: Australia, Japan · 31 Dec 2026: South Korea · 30 Sep 2027: rest of world	Same page. **If correct, India (rest of world) would still be on the older structure until the 2027 date** – confirm.
Programs	Apps Experience / Games Level Up programs reported to give lower rates from 30 Sep 2026 in AU, EEA, JP, UK, US	Play Console Help

Effect on the sample plans (illustrative)

Contribution if the same plan were sold in-app, replacing the gateway fee with a store commission on the full price. Uses the same variable-cost assumptions as the financial model; ignores tax handling differences, which vary by store and country.

Plan	Price	Contribution via own gateway	In-app at 30%	In-app at 15%	Impact
Starter	₹999	₹529	₹248	₹398	13% lower at 15% vs web
Professional	₹2,499	₹1,757	₹1,054	₹1,429	13% lower at 15% vs web
Business	₹5,999	₹4,310	₹2,624	₹3,524	13% lower at 15% vs web

Decision rules

1. Prefer web billing for B2B subscriptions where the store's rules allow it; confirm the current rules for your category.
2. If in-app sales are required, price the in-app plan so contribution stays ≥ the web plan's margin target, or accept a lower margin knowingly.
3. Enrol in Apple's Small Business Program as soon as eligible; check Google programs for your region.
4. Budget fixed store costs: Apple USD 99/year (local price at enrolment) and Google USD 25 one-time (to confirm). Add them to the one-time/annual cost lines when mobile apps are in scope.
5. Re-verify all store fees at every G7 and G8 review.

Rollout plan – first 150 days

When	Focus	Key actions	Owners
Days 1–30	Foundation	Fill intake; set up company GitHub org, registrar, password manager, MFA everywhere; adopt templates; create dashboard Project 0 backlog; start Stage 1 for first product	Owner, Tech Lead
Days 31–60	First product to G3	Complete G1–G3; choose stack; start Stage 4 designs; build cost model with real provider prices	Product Owner, Tech Lead, Designer
Days 61–90	Build and cost	G4–G5; weekly demos; start G6; approve cost and pricing (G7–G8); line up 3 pilot customers	All
Days 91–150	Launch	Finish G6; security review; G9 deploy; pilots live; maintenance agreement (G11)	DevOps, QA, Support
Days 151+	Sell and operate	G10 commercial approvals; monthly maintenance report; review CAC, churn, margin; iterate roadmap through change control	Sales, Support, Owner

Durations are planning guides for a small team and should be recalibrated from your own timesheets after the first project.

Worked example – SlotWise end to end

Product: SlotWise – online appointment booking, reminders and no-show reduction for small clinics in India. **Status: worked example for training.** Market figures below are illustrative placeholders that demonstrate the method; they are not market facts and the competitor research has deliberately not been invented.

S1. Market research (method + placeholders)

Item	Worked example
Segment / ICP	Independent clinics with 1–5 doctors, a receptionist, phone-based booking today
Problem	Missed calls, double bookings, 15–30% no-shows (ASSUMPTION – must be validated in interviews)
Buyer / user / decider	Buyer & decider: clinic owner-doctor; users: receptionist and patients
Competitor list to research (≥5)	*Researcher to fill:* name, URL, pricing page, features, reviews, access date. Candidate search routes: G2/Capterra category 'Medical scheduling', Product Hunt, Google Trends, Reddit r/india, clinic-owner groups.
TAM formula	N clinics in India × ACV. ACV = ARPA ₹2,274 × 12 = ₹27,288. N must come from a government/industry source – **do not guess**.
SAM formula	N reachable (target cities, 1–5 doctors, smartphone-using) × ACV
SOM formula	Tenants winnable in 3 years = sales capacity × conversion; with 1 sales FTE closing 4 per month ≈ 48/yr → ~100–150 in 3 years after churn
Recurring revenue	Subscription tiers, WhatsApp reminder bundle, onboarding fee, premium support

> **G1 decision rule**
> Proceed only if ≥3 of 10 clinic owners rank missed appointments among top-3 problems and ≥3 agree to a paid pilot at ≥ ₹999/month.

S2. PRD summary and I/O freeze (excerpt)

Module	Key features	Roles
Booking	Public booking page, slot rules, buffer times, holidays, double-booking prevention	Patient, Receptionist, Doctor
Reminders	Email + SMS/WhatsApp reminders T-24h/T-2h, confirm/cancel link	System, Receptionist
Patients	Minimal patient record (name, phone, email), consent flag	Receptionist, Doctor
Billing	Subscription via payment gateway; invoices with GST	Owner
Reports	Daily schedule PDF, no-show rate CSV	Owner, Doctor
Admin	Tenant management, plan limits, audit log	Platform admin

Input specification (excerpt)

Field	Type	Unit	Req.	Default	Validation	Min/Max	Source	DB mapping
slot_duration	integer	minutes	Yes	15	multiple of 5	5 / 120	Doctor settings form	doctors.slot_minutes
patient_phone	string	E.164	Yes	–	regex ^\+[1-9]\d{7,14}$	8 / 15 digits	Booking form	patients.phone
appointment_start	timestamptz	UTC (shown in Asia/Kolkata)	Yes	–	future; inside working hours; slot free	now / +180 days	Booking form	appointments.start_at
consent_reminders	boolean	–	Yes	false	must be true to send reminders	–	Booking form	patients.consent_reminders

Output specification (excerpt)

Output	Format	Calculation	Precision	Display	Export	Error handling
No-show rate	percentage	no_shows ÷ (completed + no_shows) in period	1 decimal	Reports dashboard	CSV, PDF	Return 0.0% with note if denominator is 0
Daily schedule	list	appointments for doctor/day ordered by start	–	Doctor dashboard	PDF	Empty-state message if none

Sample scenario + acceptance criterion

Input: slot_duration 15, doctor works 09:00–13:00, patient requests 10:15 on a free day. **Output:** appointment created 10:15–10:30; reminders queued for T-24h and T-2h. **Given/When/Then:** Given slot 10:15 is already booked, when another patient requests it, then the API returns 409 and the UI offers the next three free slots.

S3. Architecture decision

Criterion (weight)	A: Next.js + Node/TS modular monolith + PostgreSQL on PaaS	B: React SPA + FastAPI + PostgreSQL on a cloud VM
Development cost (20%)	4 – one language end-to-end	3 – two languages
Hosting/ops cost (15%)	4 – managed, minimal ops	3 – VM patching is ours
Scalability (10%)	3 – fine to ~5,000 users per model	4
Security (15%)	4 – managed TLS, patching	3 – more surface to own
Performance (10%)	4	4
Developer availability (10%)	5	4
Vendor lock-in (10%)	3 – PaaS-specific features avoided	5
Long-term operating cost (10%)	3 – PaaS premium at scale	4
Weighted score	**3.80**	**3.60**

Decision (illustrative): Option A; modular monolith (modules: auth, booking, reminders, billing, reports, admin) with background worker for reminders; PostgreSQL; object storage for exports. Microservices rejected – no independent scaling need. Threat-model highlights: tenant isolation (row-level checks on every query), phone/email PII protection, webhook signature verification, rate limiting on booking page.

S4. Repository setup

```
slotwise/
├─ apps/web  apps/api
├─ packages/ui  packages/shared
├─ docs/{requirements,architecture,database,api,deployment,testing}
├─ tests/{unit,integration,e2e}
├─ infra/
├─ .github/{workflows,ISSUE_TEMPLATE,pull_request_template.md}
├─ .env.example  .gitignore  README.md  CONTRIBUTING.md
```

S5. Production cost (computed)

See section 'Financial Model'. Expected development labour for SlotWise is **₹10,88,000** (optimistic ₹7,41,000, pessimistic ₹15,83,000) over 1270 expected hours.

S6. Pricing

ARPA ₹2,274, contribution per tenant ₹1,526, break-even **28 tenants**, LTV:CAC 6.4. Full plan table in 'Financial Model'.

S7. Deployment record (template filled with PLACEHOLDERS)

Step	Value	Status
Domain	[to be checked at registrar – availability NOT assumed]	Pending
DNS	[exact records copied from hosting/email provider]	Pending
TLS	Provider-issued certificate; HTTPS redirect on	Pending
Migrations	Run on production before traffic switch	Pending
Backup restore test	Restored to staging, row counts match	Pending
Smoke tests	Sign-up, book, remind, pay, export	Pending
Rollback test	Previous release re-deployed < 15 min	Pending

> **Honesty rule**
> No claim of 'production-ready' is made for the sample: every deployment row is Pending until evidence is attached.

S8. Customer delivery

1. Pilot with 3 clinics for 60 days at 50% price; written consent for case study.
2. SOW covering: configuration, data import, training (2 × 90 min), acceptance tests.
3. Subscription terms: monthly/annual billing, fair-use limits, data export on exit within 30 days, data deletion within 90 days, customer owns its data; company owns platform code and reusable components; third-party licences remain with licensors.
4. Handover pack: user manual, admin manual, support contacts and hours, escalation matrix.

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The 20 standard templates

#	Template	Purpose	Mandatory fields
1	Market Research Report	Opportunity evidence	Segment, problem, TAM/SAM/SOM workbook, sources with access dates, recommendation, kill criteria
2	Competitor Analysis	Compare 5–10 competitors	Name, URL, access date, pricing, features, tech signals, reviews, positioning, gaps
3	Customer Discovery	Interview log	Interviewee role, date, current workflow, pain rank, spend today, willingness to pay, quotes, consent
4	PRD	Product definition	Purpose, customers, roles/permissions, modules, journeys, NFRs, backup, maintenance
5	Input/Output Specification	Data contract	Per field: type, unit, required, default, validation, min/max, source, DB map; per output: format, logic, precision, display, export, API shape, errors
6	Scope Freeze	Baseline + change control	Version, approvers, out-of-scope, CR log (ID, justification, tech/cost/timeline/security impact, approvals)
7	Architecture Document	Technical design	Options scored, diagrams, ERD, API, auth, threat model, infrastructure, ADRs
8	UI/UX Approval	Design sign-off	Sitemap, flows, hi-fi links, design tokens, usability findings, approvals
9	GitHub Setup	Repo checklist	Repo URL, branch rules, CI checks, secrets storage, environments, release tags
10	Development Report	Weekly progress	Modules done, requirement refs, hours vs estimate, demo notes, blockers
11	Code Review	PR review record	Scope, correctness, security, tests, readability, decision
12	QA Report	Test evidence	Test plan, cases run/passed/failed, defects by severity, regression, UAT sign-off
13	Security Report	Pre-launch review	ASVS checklist, dependency scan, secrets scan, auth/RBAC tests, findings, residual risk
14	Cost Estimation	Phase 7 workbook	Labour three-point, setup, monthly, capacity, three models, source URLs + dates
15	Pricing Model	Phase 8 workbook	Three pricing approaches, plans, margins, break-even, CAC/LTV, assumptions
16	Domain/DNS Report	Domain record	Registrar, owner, MFA, renewal date, DNS records (exact values), propagation proof
17	Deployment Report	Release record	Build, env vars (names only), migrations, smoke tests, rollback proof, approver
18	Customer Showcase	Demo pack	Script, dataset, limitations list, pricing, pilot proposal, next steps
19	Project Handover	Delivery close	Deliverables, credentials transfer procedure, manuals, training, acceptance form
20	Maintenance Report	Monthly ops	Uptime, incidents, patches, backup test, cost vs budget, support stats, actions

Each template is stored in the dashboard Documents module, versioned, and linked to its gate. The Change Request, Weekly Status and Risk Register forms are below.

Standard forms

Change Request form

Field	Content
Change Request ID	CR--
Requested by / date	
Description	
Business justification	
Technical impact	
Cost impact (₹)	
Timeline impact (days)	
Security impact	
Customer approval	Name / date / signature
Product owner approval	Name / date / signature
Decision	Approved / Rejected / Deferred

Project record (14 mandatory fields)

Project record field	Value
Project ID	
Product owner	

Project record field	Value
Technical lead	
Assigned developers	
QA owner	
Customer contact	
Budget	
Delivery timeline	
Approved scope (version)	
Production cost estimate	
Sales price	
Expected margin	
Current stage / gate	
Risk register link	

Weekly employee report

Weekly status field	Content
Employee / week	
Completed tasks	
Pending tasks	
Blockers	
Time spent (h) by project ID	
Quality issues	
Upcoming deliverables	

Risk register (example rows)

ID	Risk	Likelihood (1–5)	Impact (1–5)	Score	Mitigation	Owner	Status
R1	Key developer unavailable	3	4	12	Cross-training, docs	Tech Lead	Open
R2	Provider price change	3	3	9	Re-run cost model each quarter	Finance	Open
R3	Scope creep	4	4	16	Change-request gate	Product Owner	Open
R4	Security incident	2	5	10	ASVS review, backups, incident drill	Tech Lead	Open

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Internal management dashboard specification

Modules

#	Module	Purpose	Primary roles
1	Company Overview	KPIs, alerts, pending approvals	All
2	Project Management	Project record, stage tracker, timeline	PM, Tech Lead
3	Market Research	Reports, competitor sheets, sources	Product
4	Product Requirements	PRD/I-O versions	Product
5	Scope Approval	Freeze + CR workflow	Product, Owner
6	Architecture Approval	Docs + ADR + approval	Tech Lead
7	UI/UX Approval	Design links + sign-off	Design, Product
8	Employee Assignments	Roster, capacity	PM
9	Development Tracking	Tasks, hours vs estimate	Dev
10	GitHub Repository Tracking	Repos, PRs, CI status, releases	Tech Lead
11	QA & Testing	Test runs, defects	QA
12	Production Cost Calculator	Phase 7 model	Finance
13	SaaS Pricing Calculator	Phase 8 model	Finance, Owner
14	Domain & DNS Tracker	Domains, renewals, records	DevOps
15	Deployment Tracker	Releases, rollbacks	DevOps
16	Customer Management	Accounts, contacts, consents	Sales, Support
17	Sales Pipeline	Stages, forecasts	Sales
18	Subscription Management	Plans, renewals, MRR	Finance, Support
19	Maintenance Management	Tickets, incidents, SLA	Support, DevOps
20	Reports & Analytics	Exports, trend charts	Managers

Management KPIs

KPI	Definition
Total / active / delayed projects	Count from project records; delayed = current date > planned date of current stage
Project budgets vs actual expenditure	Budget field vs timesheet hours × loaded rate + invoices
Estimated vs actual production cost	Phase 7 model vs provider invoices
Expected revenue	Σ weighted pipeline value (stage probability × ACV)
MRR	Σ active subscription price (ex-GST, net of discounts)
Gross margin	(Revenue − cost of revenue) ÷ Revenue; cost of revenue defined in pricing model
Employee productivity	Delivered story points ÷ hours; never used alone for appraisal
Customer satisfaction	CSAT/NPS from post-onboarding and quarterly surveys
Pending approvals	Open gate/CR requests per approver

Roles and permissions

Role	Can do
Employee	Update tasks, log time, upload deliverables, request approvals
Lead	Review, approve technical gates, assign work
Manager / Owner	Approve, reject or request changes at all gates; see finance
Finance	Edit cost/pricing models, view revenue
Admin	Users, roles, audit log

Each project shows a stage tracker G1 → G12. Dashboard is responsive, role-based (server-side checks), and audit-logged. It is built as Project 0 using this same operating system.

Incidents, maintenance agreement and DNS

Incident matrix

Level	Definition	Owner	Response target*	Escalation	Communication	Recovery
Critical	Entire service down or confirmed serious security incident	Tech Lead (incident commander)	Acknowledge ≤ 30 min, 24×7 only if staffed	Managing Director immediately	Status page + email to affected customers; regulator/customer notification per law/contract	Rollback or failover; preserve evidence; post-incident review ≤ 5 days
High	Major business function unavailable	DevOps on-call	≤ 2 business hours	Tech Lead after 2 h	Email update every 4 h	Hotfix branch, expedited review
Medium	Limited functionality affected, workaround exists	Support Lead → Developer	≤ 1 business day	Tech Lead if > 3 days	Ticket updates	Fix in next release
Low	Minor or cosmetic	Support Lead	≤ 3 business days	None	Ticket	Backlog

Maintenance agreement outline

Maintenance plan item	Starter	Business
Service hours	Mon–Fri 10:00–18:00 IST	Mon–Sat 09:00–20:00 IST
Channels	Email	Email + chat + phone
Included work	Bug fixes, patching, backups	Plus ≤ 8 h/month minor changes
Exclusions	New features, customer-caused data issues, third-party outages, force majeure	Same
Price	Included in subscription	Included / add-on per quote

*Targets are examples. Never publish an SLA (uptime %, response times) until on-call staffing, monitoring and architecture can demonstrably meet it.

DNS record reference

Record	Purpose	Rule
A / AAAA	Maps name to IPv4 / IPv6 address	Use exact values from hosting provider
CNAME	Alias to another hostname	Cannot coexist with other records on same name; apex needs ALIAS/flattening if supported
MX	Mail routing	Copy from email provider; never overwrite existing MX
TXT	Verification, SPF, DKIM, DMARC	Only one SPF record per domain; merge includes
SPF	Authorised sending servers	Start with provider include; end with ~all then -all after testing
DKIM	Message signing	Publish selector record exactly as given
DMARC	Policy and reporting	Begin p=none with rua reporting; move to quarantine/reject after review
CAA	Restricts certificate issuers	Must include the CA your host uses

Assumption register

ID	Assumption	Where used	Replace with
A1	Fully loaded hourly rates	Labour cost	HR/finance actuals
A2	Hours per activity (O/M/P)	Labour cost	Past project timesheets
A3	Provider unit prices	Infrastructure model	Official pricing pages + dates
A4	Usage per user (DAU, requests, MB)	Capacity	Analytics from pilot
A5	Plan prices & mix	Pricing	Customer discovery + competitor pages
A6	Churn, refunds, failed-payment rates	LTV, profit	Billing data after 3 months
A7	Gateway fee	Contribution	Gateway contract
A8	GST 18%, corporate tax 25.17%	Pricing, profit	CA / CBIC / Income-tax dept
A9	CAC ₹6,000	LTV:CAC	Actual sales & marketing spend

Source index

Source	URL	Verified on (fill)
VS Code	https://code.visualstudio.com/docs	___/___/_____
GitHub	https://docs.github.com/	___/___/_____
Next.js	https://nextjs.org/docs	___/___/_____
React	https://react.dev/	___/___/_____
Node.js	https://nodejs.org/docs/	___/___/_____
PostgreSQL	https://www.postgresql.org/docs/	___/___/_____
Vercel	https://vercel.com/docs	___/___/_____
Cloudflare	https://developers.cloudflare.com/	___/___/_____
AWS	https://docs.aws.amazon.com/	___/___/_____
Google Cloud	https://cloud.google.com/docs	___/___/_____
Microsoft Azure	https://learn.microsoft.com/azure/	___/___/_____
OWASP	https://owasp.org/	___/___/_____
OWASP ASVS	https://asvs.owasp.org/	___/___/_____
NIST SSDF SP 800-218	https://csrc.nist.gov/pubs/sp/800/218/final	___/___/_____
ICANN	https://www.icann.org/	___/___/_____
MeitY	https://www.meity.gov.in/	___/___/_____
CBIC (GST)	https://cbic-gst.gov.in/	___/___/_____
WCAG 2.2	https://www.w3.org/TR/WCAG22/	___/___/_____
Google SRE book	https://sre.google/sre-book/table-of-contents/	___/___/_____
Apple Developer Program – what's included	https://developer.apple.com/programs/whats-included/	**2026-11-03** – page fetched
Apple App Store Small Business Program	https://developer.apple.com/app-store/small-business-program/	**2026-11-03** – page fetched
Google Play service fees	https://support.google.com/googleplay/android-developer/answer/112622	2026-11-03 – search summary only; page blocked, re-verify
Google Play lower service fees (2026)	https://support.google.com/googleplay/android-developer/answer/16954621	2026-11-03 – search summary only; page blocked, re-verify
Android Developers Blog – expanded billing	https://android-developers.googleblog.com/2026/06/play-expanded-billing.html	2026-11-03 – search summary only; re-verify

Verification required
Links are authoritative starting points. Pricing, tax rates and regulations change; whoever uses a figure must open the page, confirm it, and record the date. Only the two Apple pages marked above were fetched directly; Google Play figures are search-reported and flagged as such; no cloud-provider price is quoted as fact.

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